RESULTS — TMUA 2017 P2

18 / 20

20 attempted, 0 left blank · total time on paper 1:14:39

← Dashboard | Exam simulator

TMUA SCORE

9.0 2024+ converted 9.0 Official

Official: TMUA 2017's published score conversion. 2024+ converted: our estimate of the equivalent on today's scale.

BY TOPIC

Area Integration	1 / 1
Combinatorics	0 / 1
Differentiation	3 / 3
Exponentials and Logar…	2 / 2
General Algebra	2 / 2
General Trigonometry	1 / 2
Geometry	1 / 1
Graphs of Functions	0 / 1
Integration	2 / 2
Logic Counterexample	2 / 2
Logic Deduction	5 / 6
Logic Negation	1 / 1
Logic Sufficiency	2 / 2
Sequences and Series	3 / 3
Transformation of Grap…	1 / 1

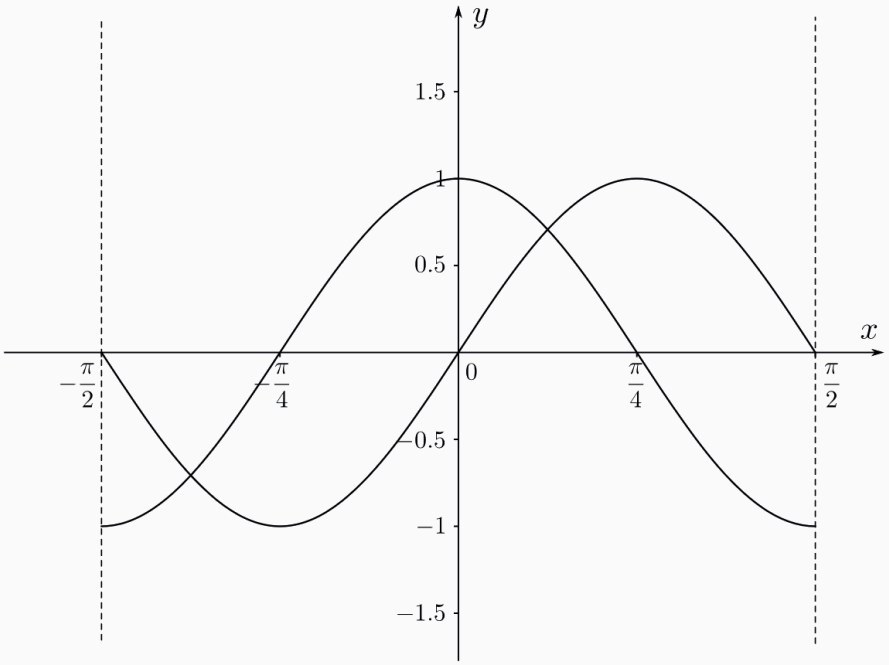
BY DIFFICULTY

3.6	1 / 1
4.3	1 / 1
4.6	1 / 1
4.7	1 / 1
4.8	1 / 1
5.0	2 / 2
5.4	1 / 1
5.6	1 / 1
5.8	2 / 2
6.0	1 / 2
6.2	1 / 1
6.3	1 / 1
6.4	0 / 1
6.5	1 / 1
6.6	1 / 1
6.7	2 / 2

Per-question breakdown

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	A	A	✓	1:51	Bookmark	Review
2	E	E	✓	2:55	Bookmark	Review
3	G	G	✓	4:10	Bookmark	Review
4	B	B	✓	2:50	Bookmark	Review
5	B	B	✓	2:44	Bookmark	Review
6	A	A	✓	3:05	Bookmark	Review
7	E	E	✓	6:19	Bookmark	Review
8	E	E	✓	3:47	Bookmark	Review
9	D	D	✓	1:32	Bookmark	Review
10	D	D	✓	7:51	Bookmark	Review
11	B	B	✓	2:51	Bookmark	Review
12	E	C	✗	6:33	Bookmark	Review
13	B	B	✓	5:11	Bookmark	Review
14	F	F	✓	1:41	Bookmark	Review
15	D	D	✓	6:28	Bookmark	Review
16	C	C	✓	5:15	Bookmark	Review
17	F	F	✓	2:32	Bookmark	Review
18	B	B	✓	0:35	Bookmark	Review
19	E	E	✓	3:21	Bookmark	Review
20	C	B	✗	3:08	Bookmark	Review

QUESTION



The diagram shows the graphs of $y = \sin 2x$ and $y = \cos 2x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Which one of the following is **not** true?

A $\cos 2x < \sin 2x < \tan x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

B $\cos 2x < \tan x < \sin 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

C $\sin 2x < \cos 2x < \tan x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$ CORRECT

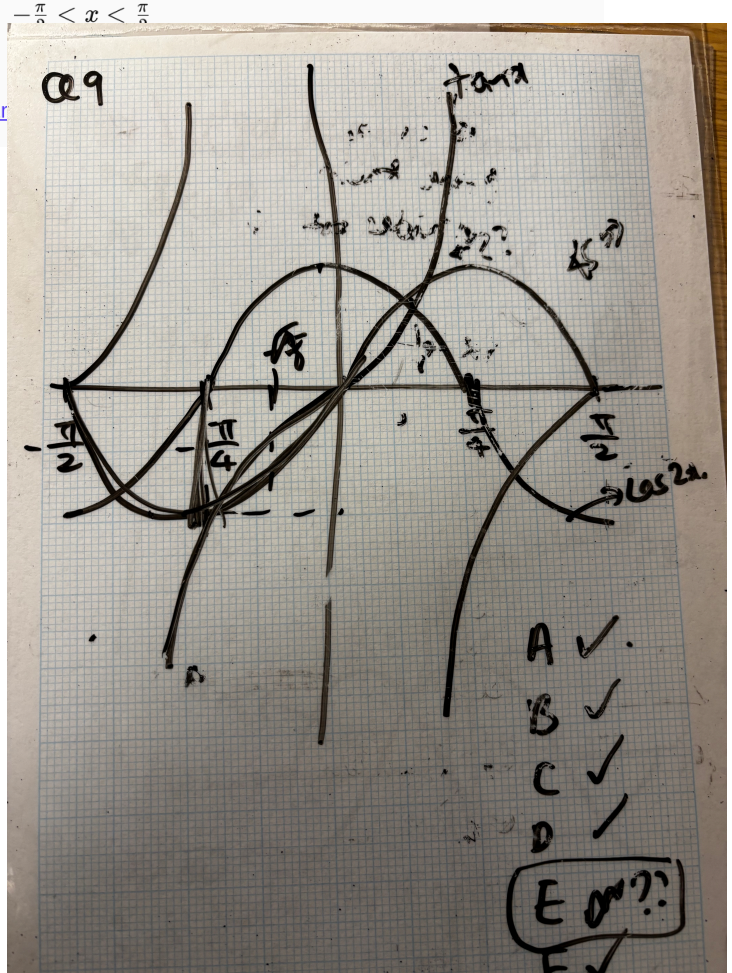
D $\sin 2x < \tan x < \cos 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

E $\tan x < \sin 2x < \cos 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$ YOUR CHOICE

F $\tan x < \cos 2x < \sin 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

SOLUTION

The correct solution is option C. For details see UAT official worked ar



20	C	B	✕	3:08	Bookmark	Hide

QUESTION

I have forgotten my 5-character computer password, but I know that it consists of the letters a, b, c, d, e in some order. When I enter a potential password into the computer, it tells me exactly how many of the letters are in the correct position.

When I enter abcde, it tells me that none of the letters are in the correct position. The same happens when I enter cdbea and eadbc.

Using the best strategy, how many **further** attempts must I make in order to **guarantee** that I can **deduce** the correct password?

A None: I can deduce it immediately

B One — CORRECT

C Two — YOUR CHOICE

D Three

E More than three

SOLUTION

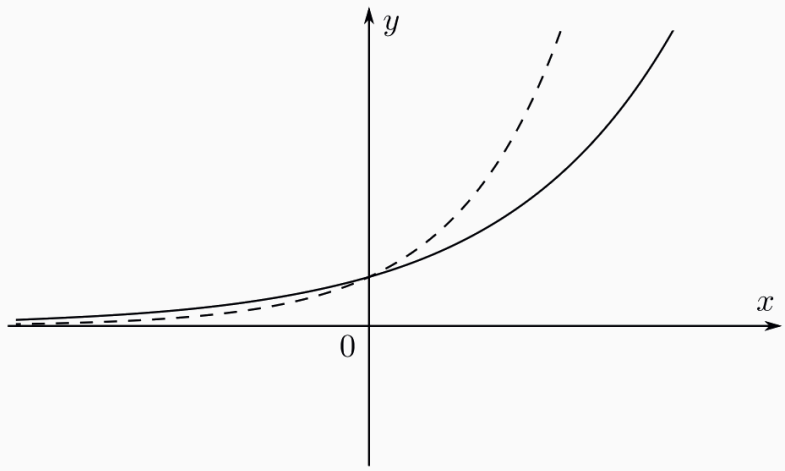
The correct solution is option B. For details see UAT official worked answers

Logic Deduction | Combinatorics | Difficulty 6.4

Question that took very long:

7 E E ✓ 6:19 Bookmark Hide

QUESTION



The graphs of two functions are shown here:

- $y = a^x$ is shown with a solid line, where a is a positive real number
- $y = f(x)$ is shown with a dashed line

Which of the following statements (**1**, **2**, **3**, **4**) could be true?

1 $f(x) = b^x$ for some $b > a$
2 $f(x) = b^x$ for some $b < a$
3 $f(x) = a^{kx}$ for some $k > 1$
4 $f(x) = a^{kx}$ for some $k < 1$

A	**1** only	**B**	**2** only
C	**3** only	**D**	**4** only
E	**1** and **3** only CORRECT	**F**	**1** and **4** only
G	**2** and **3** only	**H**	**2** and **4** only

SOLUTION

13 B B ✓ 5:11 Bookmark Hide

QUESTION

The positive real numbers $a \times 10^{-3}$, $b \times 10^{-2}$ and $c \times 10^{-1}$ are each in standard form, and

$$(a \times 10^{-3}) + (b \times 10^{-2}) = (c \times 10^{-1}).$$

Which of the following statements (I, II, III, IV) **must** be true?

I $a > 9$
II $b > 9$
III $a < c$
IV $b < c$

A	I only	**B**	II only CORRECT
C	I and II only	**D**	I and III only
E	I and IV only	**F**	II and III only
G	II and IV only	**H**	I, II, III and IV

SOLUTION

The correct solution is option B. For details see UAT official worked answers

Logic Deduction | General Algebra | Difficulty 6.6

11 B B ✓ 2:51 Bookmark Hide

QUESTION

The function $f(x)$ is increasing and $f(0) = 0$.

The positive constants a and b are such that $a < b$.

The area of the region enclosed by the curve $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$ is denoted by R.

The function $g(x)$ is defined by $g(x) = f(x) + 2f(b)$.

Which of the following is an expression for the area enclosed by the curve $y = g(x)$, the x-axis and the lines $x = a$ and $x = b$?

A $R + (b - a)f(b)$

B $R + 2(b - a)f(b)$ CORRECT

C $R + 2f(b) - f(a)$

D $R + 2f(b)$

E $R + (f(b))^2$

F $R + (f(b))^2 - (f(a))^2$

G $R + 2(f(b) - f(a))f(b)$

SOLUTION

The correct solution is option B. For details see UAT official worked answers

Area Integration Difficulty 5.4

10 $f(x)$ is a function defined for all real values of x.

Which one of the following is a **sufficient** condition for $\int_1^3 f(x)\,dx = 0$?

~~A $f(2) = 0$~~

~~B $f(1) = f(3) = 0$~~

~~C $f(-x) = -f(x)$ for all x~~ ⇒ odd function

⇒ not 'sufficient'

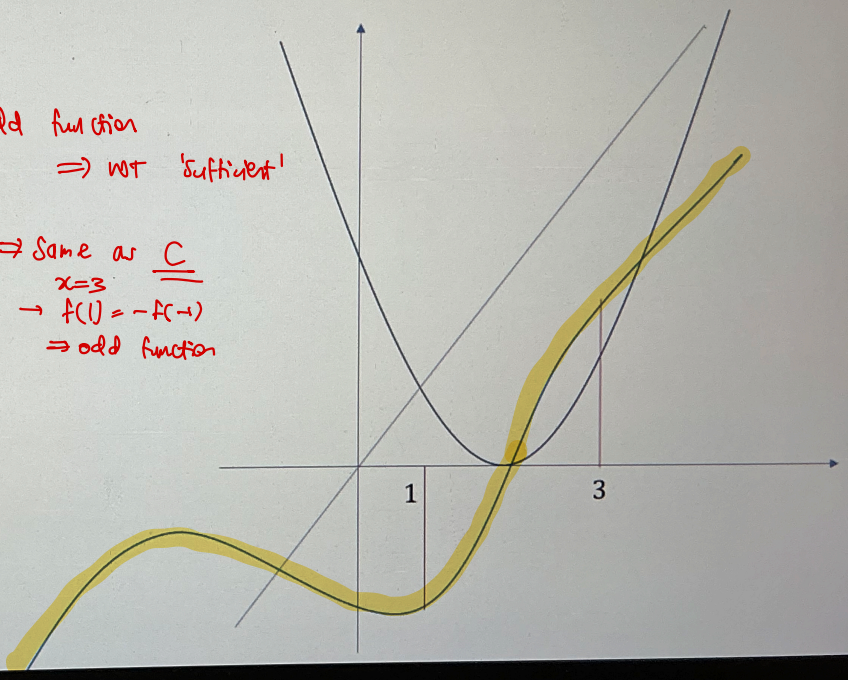
D $f(x+2) = -f(2-x)$ for all x

~~E $f(x-2) = -f(2-x)$ for all x~~ ⇒ same as C

$x=3$

→ $f(1) = -f(-1)$

⇒ odd function



needs to have

rotational symmetry

11 The function $f(x)$ is increasing and $f(0) = 0$.

The positive constants a and b are such that $a < b$.

The area of the region enclosed by the curve $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$ is denoted by R.

The function $g(x)$ is defined by $g(x) = f(x) + 2f(b)$.

Which of the following is an expression for the area enclosed by the curve $y = g(x)$, the x-axis and the lines $x = a$ and $x = b$?

A $R + (b - a)f(b)$

B $R + 2(b - a)f(b)$

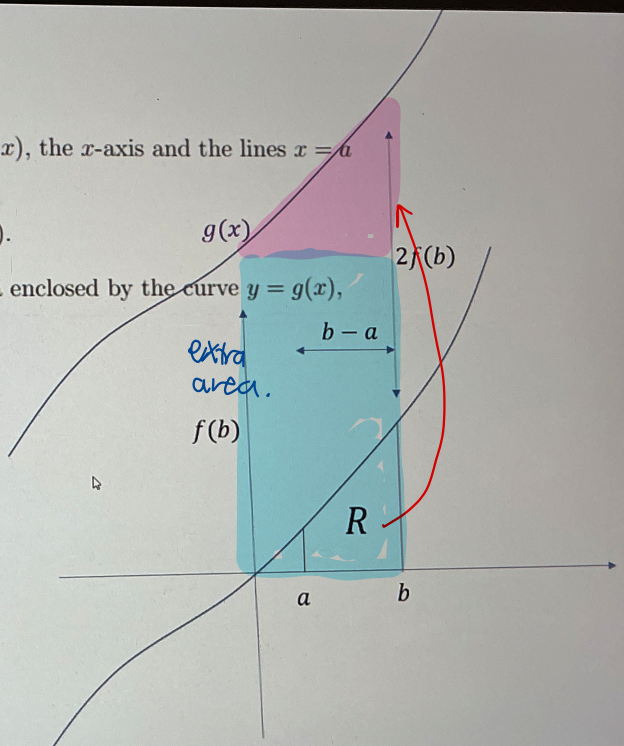
C $R + 2f(b) - f(a)$

D $R + 2f(b)$

E $R + (f(b))^2$

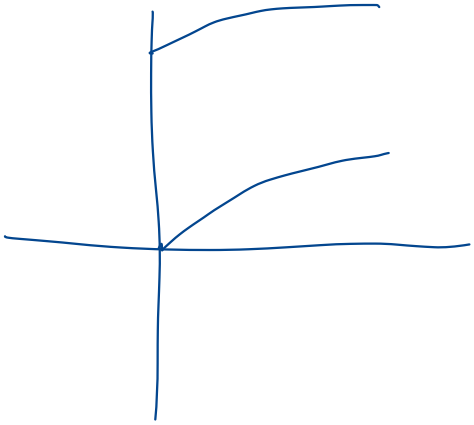
F $R + (f(b))^2 - (f(a))^2$

G $R + 2(f(b) - f(a))f(b)$



graph shifted up for every value between a & b

~~add~~

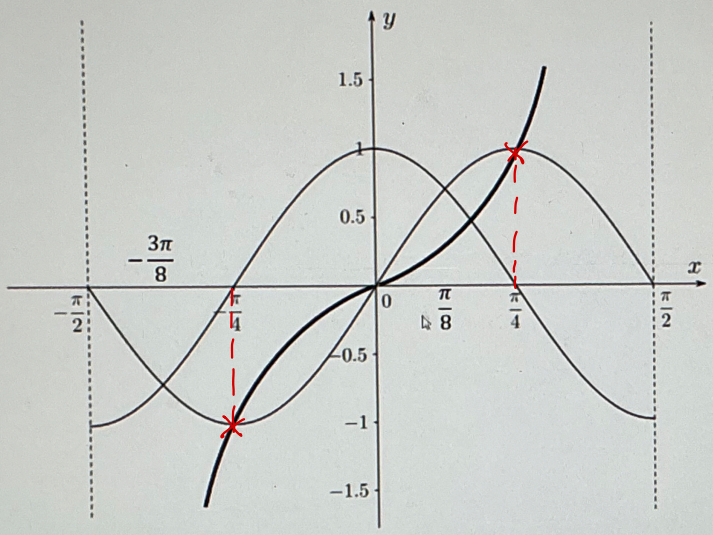


$$\int_a^b f(x) + 2f(b)$$

$$= \int_a^b f(x) + 2\int_a^b f(b)$$

$$= R + 2$$

12 The diagram shows the graphs of $y = \sin 2x$ and $y = \cos 2x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$



Which one of the following is **not** true?

A $\cos 2x < \sin 2x < \tan x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

B $\cos 2x < \tan x < \sin 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

C $\sin 2x < \cos 2x < \tan x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

D $\sin 2x < \tan x < \cos 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

E $\tan x < \sin 2x < \cos 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

F $\tan x < \cos 2x < \sin 2x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

★★★ it's tan(x)★ NOT tan2x

★★ NEED TO READ THE Q very carefully.

13 The positive real numbers $a \times 10^{-3}$, $b \times 10^{-2}$ and $c \times 10^{-1}$ are each in standard form, and

$$(a \times 10^{-3}) + (b \times 10^{-2}) = (c \times 10^{-1}).$$

Which of the following statements (I, II, III, IV) **must** be true?

I $a > 9$

II $b > 9$

III $a < c$

IV $b < c$

A I only

B II only

C I and II only

D I and III only

E I and IV only

F II and III only

G II and IV only

H I, II, III and IV

Come up with counter-examples

R2 drew2's solution

$$\frac{a}{1000} + \frac{b}{100} = \frac{c}{10}$$

$\therefore$ $a + 10b = 100c$

Standard form $\therefore$ $0 < a, b, c < 10$